Rockchip RK3399 PCIe Developer Guide

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Preface

Overview

Product Versions

Chip Name	Kernel Version
RK3399	4.4, 4.19

Intended Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Lin Tao	2017-11-25	Initial version
V1.0.1	Huang Ying	2021-04-28	Updated format; renamed file to support RK3399
V1.1.0	Lin Tao	2021-06-10	Removed bandwidth description; updated Lane interleaving description

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1. DTS Configuration

1. `ep-gpios = <&gpio3 13 GPIO_ACTIVE_HIGH>;`

This item configures the PERST# reset signal for the PCIe interface. Whether using a slot-mounted or soldered device, locate this pin on the schematic and configure it correctly.

Failure to do so will prevent link establishment.

2. `num-lanes = <4>;`

This configuration sets the number of lanes used by the PCIe device. Adjustment is typically unnecessary as the software can automatically detect and disable unused lanes to save power.

3. `status = <okay>;`

This configuration must be enabled in both the pcie0 and pcie_phy nodes. It is disabled by default because, in the absence of peripherals, PCIe initialization incurs a significant detection delay, unnecessarily increasing boot time. Therefore, projects requiring PCIe should enable this feature manually.

4. `vpcie3v3-supply = <&vdd_pcie3v3>;`

This configuration is optional and is used to configure the 3V3 power supply for PCIe peripherals. If board-level control of the 3V3 enable signal for PCIe peripherals is required, define a corresponding set of regulators as shown in the example. For regulator configuration details, refer to Documentation/devicetree/bindings/regulator/.

5. `vpcie1v8-supply = <&vdd_pcie1v8>;`

Refer to item 4.

6. `vpcie0v9-supply = <&vdd_pcie1v8>;`

Refer to item 4.

2. menuconfig Configuration

1. The following configurations must be enabled to correctly use PCIe-related features.

```
CONFIG_PCI=y
CONFIG_PCI_DOMAINS=y
CONFIG_PCI_DOMAINS_GENERIC=y
CONFIG_PCI_SYSCALL=y
CONFIG_PCI_BUS_ADDR_T_64BIT=y
CONFIG_PCI_MSI=y
CONFIG_PCI_MSI_IRQ_DOMAIN=y
CONFIG_PHY_ROCKCHIP_PCIE=y
CONFIG_PCIE_ROCKCHIP=y
CONFIG_PCIEPORTBUS=y
CONFIG_PCIEASPM=y
CONFIG_PCIEASPM_POWERSAVE=y
CONFIG_PCIE_PME=y
```

```
CONFIG_GENERIC_MSI_IRQ=y
CONFIG_GENERIC_MSI_IRQ_DOMAIN=y
CONFIG_IRQ_DOMAIN=y
CONFIG_IRQ_DOMAIN_HIERARCHY=y
```

2. Enable NVMe devices (SSDs based on the PCIe interface)

```
CONFIG_BLK_DEV_NVME=y
```

3. Enable the AHCI device (SSD converted from PCIe to SATA)

```
CONFIG_SATA_PMP=y
CONFIG_SATA_AHCI=y
CONFIG_SATA_AHCI_PLATFORM=y
CONFIG_ATA_SFF=y
CONFIG_ATA=y
```

Note: By default, the open-source kernel version 4.4 only supports devices listed in drivers/ata/ahci.c. For devices beyond this list, please contact the manufacturer or authorized distributor for support.

4. Enabling PCIe Interface for WIFI

Please refer to the respective vendor documentation for WIFI configuration.

3. cmdline Configuration

For detailed explanations, refer to the kernel documentation Documentation/kernel-parameters.txt. Only key parameters are listed and explained here.

1. nomsi

To use Legacy Interrupt mode, add pci=nomsi to the parameters.

2. pcie_bus_safe

To adjust the Maximum Payload Size (MPS) of all devices across PCIe hierarchy levels to the maximum value, configure pci=pcie_bus_safe. This can improve bandwidth.

3. pcie_aspm

To disable dynamic power management of the PCIe link for testing purposes, configure pcie_aspm=off. Otherwise, it will be automatically configured based on hardware negotiation by default.

4. Common Application Issues

Q1: Customers find routing difficult. Can lanes be interleaved between different lanes?

A1: Theoretically, interleaving is possible. The RC's lane[1-4] can correspond arbitrarily to the EP/switch's lane[1-4]. No software modifications are required. However, this has not been practically verified by our company. Customers should evaluate this carefully.

Q2: Can differential signals within the same lane be interleaved? For example, can RC's lane1 RX+ correspond to EP/Switch's RX-, and TX+ correspond to EP/Switch's TX-? Or can RX positive/negative and TX positive/negative be cross-connected? How should such cases be handled?

A2: Connections can be made arbitrarily; no additional software handling is required. The PCIe detection state machine already accounts for all such scenarios.

Q3: The RK3399 has only one Root Complex (RC) but four lanes. Can these four lanes be split, for example into four 1-1-1-1 configurations, two 2-2 configurations, or other combinations?

A3: The RK3399 does not support such requirements. If customers need to connect multiple devices, a PCIe switch should be used. We have currently debugged Pericom switches, which are likely the most cost-effective option.

Q4: Does the RK3399 support SSDs?

A4: Please note that there are two types of SSDs: one is NVMe, which uses the PCIe bus at the physical signal layer; the other is mSATA, which uses the SATA bus. If support for the latter is required, customers should purchase a PCIe-to-SATA or USB-to-SATA adapter.

Q5: Since NVMe is supported, what is the maximum capacity, and can it be used as a boot drive?

A5: The capacity of the storage device depends on the file system; there are no limitations at the Driver level. Additionally, NVMe boot on the RK3399 is supported starting from U-Boot, which requires an extra SPI-NOR flash to store the MiniLoader, as MaskROM does not include NVMe or PCIe Drivers.

Q6: What is the bandwidth when connecting NVMe via PCIe on the RK3399, and how is it tested?

A6: Testing is performed using the fio utility with the following command (customers must compile their own fio binary with libaio statically linked):

./fio -filename=/dev/block/nvme0n1 -direct=1 -iodepth 4 -thread=1 -rw=write -ioengine=libaio -bs=1M -size=200G -numjobs=30 -runtime=60 -group_reporting -name=my

The test results are approximately as follows:

测试工具：FIO
测试平台：RK3399 EVB3/PCIe1.0×4 PayLoad:256
SSD：Samsung SM961 MZVPW256HEGL

测试项	iodepth	thread	ioengine	performance	AVG IOPS
1M顺序读	4	1	libaio	784032KB/s	
1M顺序写	4	1	libaio	792945KB/s	
4K随机写	64	6	libaio		194343
8K随机写	64	6	libaio		98200
16K随机写	64	6	libaio		49159
4K随机读	64	6	libaio		170677
8K随机读	64	6	libaio		93934
16K随机读	64	6	libaio		44893

测试工具：FIO
测试平台：RK3399 EVB3/PCIe2.0×4 PayLoad:256
SSD：Samsung SM961 MZVPW256HEGL

测试项	iodepth	thread	ioengine	performance	AVG IOPS
1M顺序读	4	1	libaio	1452.2MB/s	
1M顺序写	4	1	libaio	1039.1MB/s	
4K随机写	64	6	libaio		208538
8K随机写	64	6	libaio		105924
16K随机写	64	6	libaio		47737
4K随机读	64	6	libaio		246223
8K随机读	64	6	libaio		174749
16K随机读	64	6	libaio		80787

Q7: Does the RK3399 support discrete graphics cards?

A7: Theoretically, yes, provided an ARM version of the Driver is available. In practice, it is not supported. The reason is that the physical bus address space available for BAR is only 32 MB, whereas most discrete graphics cards exceed this memory requirement. Even using a switch for adaptation is ineffective, because the switch itself has memory requirements, and after adaptation, the PCIe bus address range requirement for the Root Complex (RC) becomes even larger.

Q8: Power supply requirements for PCIe devices

A8: Generally, four types of power supplies are provided: 0.9 V, 3.3 V, 3.3 V_AUX, and 12 V. The 0.9 V supply is primarily used for PCIe Wi-Fi devices. The vast majority of devices require the 3.3 V rail. The 3.3 V_AUX is an auxiliary power supply that must remain powered during suspend to allow the device to continue operating; for example, if Wi-Fi needs to wake up the main controller, this AUX power rail must remain always-on. The 12 V supply is used for high-power devices such as switches and graphics cards. Actual configuration should follow the requirements specified by the Endpoint (EP) or switch manufacturer.

Q9: How to dynamically enable or disable ASPM support for PCIe devices on the RK3399?

A9: Currently, the following methods are available:

Method 1: Add `pcie_aspm=off` to the kernel command line.

Method 2: Configure the `link_state` node in the console. Note that this requires `CONFIG_PCIEASPM_DEBUG=y` to be enabled in the kernel configuration.

```
# cat /sys/bus/pci/devices/0000:00:00.0/power/link_state
7
# echo 0 > /sys/bus/pci/devices/0000:00:00.0/power/link_state
```

Method 3: Disable the `CONFIG_PCIEASPM` option in the kernel configuration.

Method 4: Use the `setpci` command. For example: `setpci -s 0:0 0xd0.w=0xC00`. This method requires prior knowledge of the corresponding device's BDF (Bus/Device/Function) information.

Method 5: Enter the following command in the console: `echo performance > /sys/module/pcie_aspm/parameters/policy`.

Q10: How to view PCIe device information on the RK3399

A10: Please push the `lspci` binary to the device, add executable permissions, and then execute the `lspci` command.

Common operations include `lspci -vvv` and `lspci -t`, which are used to output device attributes and various runtime statuses, and to output the PCIe topology tree structure, respectively. For remaining parameters, please execute `lspci --help` to read the help documentation.

Q11: How to modify certain Register information of PCIe devices in user space?

A11: The `setpci` tool can be used for modification. The prerequisites are obtaining the BDF of the target device using `lspci` and understanding the corresponding Register offsets in the protocol. Please use `setpci --help` to view the corresponding help documentation. This is an advanced tool and is generally not recommended for customers unless they have sufficient understanding of the PCIe protocol.

5. Exception Troubleshooting

1. Training failure

```
rockchip-pcie f8000000.pcie: PCIe link training gen1 timeout!
rockchip-pcie: probe of f8000000.pcie failed with error -1
```

Failure cause: Training failed because the peripheral is not operational. First, verify whether the ep-gpios configuration is correct. Next, check if the peripheral power supply is present and sufficient. While 3.3V is theoretically adequate, some devices require adjustment to 3.8V or even 4V to function properly. Finally, even if the voltage is sufficient, ensure that the power delivery capacity is adequate (test using an external power supply).

2. System hang during config configuration

```
[ 0.459371] pci 0000:00:00.0: bridge configuration invalid ([bus 00-00]),
reconfiguring
[ 0.459585] pci 0000:01:00.0: reg 0x10: initial BAR value 0x00000000 invalid
[ 0.460043] pci 0000:01:00.1: reg 0x10: initial BAR value 0x00000000 invalid
[ 0.460503] pci 0000:01:00.2: reg 0x10: initial BAR value 0x00000000 invalid
[ 0.460535] pci 0000:01:00.2: reg 0x14: initial BAR value 0x00000000 invalid
[ 0.460904] Bad mode in Error handler detected, code 0xbf000002 -- SError
[ 0.460919] Internal error: Oops - bad mode: 0 [#1] PREEMPT SMP
[ 0.466658] Modules linked in:
[ 0.466938] CPU: 5 PID: 1 Comm: swapper/0 Not tainted 4.4.55 #41
[ 0.467464] Hardware name: Rockchip RK3399 Excavator Board edp (Android) (DT)
[ 0.468089] task: ffffffc0f2160000 ti: ffffffc0f2168000 task.ti:
ffffffc0f2168000
[ 0.468752] PC is at rockchip_pcie_rd_conf+0xb8/0x138
[ 0.469200] LR is at pci_bus_read_config_dword+0x78/0xbc
```

In this scenario, the device operates and enumeration completes, but the system hangs during communication. In most cases, this is caused by insufficient power supply voltage or inadequate power delivery.

3. The PLL of the RK3399 PCIe PHY fails to achieve lock.

```
[ 0.440803] rockchip-pcie f8000000.pcie: no vpcie3v3 regulator found
[ 0.440831] rockchip-pcie f8000000.pcie: no vpcie1v8 regulator found
[ 0.440854] rockchip-pcie f8000000.pcie: no vpcie0v9 regulator found
[ 1.443759] pll lock timeout!
[ 1.443826] phy phy-phy@e220.5: phy poweron failed --> -22
[ 1.443847] rockchip-pcie f8000000.pcie: fail to power on phy, err -22
[ 1.443968] rockchip-pcie: probe of f8000000.pcie failed with error -22
```

Verify that the PCIE_AVDD_0V9 and PCIE_AVDD_1V8 voltages are normal and sufficient; this rail requires LDO power supply.

4. Errors occur when RK3399 PCIe and USB 3.0 are used simultaneously.

The following log shows a system anomaly when a USB device is inserted during NVMe operation.

```
[ 2.801962] Unhandled fault: synchronous external abort (0x96000210) at
0xffffff800936401c
[ 2.801997] Internal error: : 96000210 [#1] PREEMPT SMP
[ 2.803157] Modules linked in:
[ 2.803437] CPU: 2 PID: 146 Comm: nvme Not tainted 4.4.16 #146
[ 2.803949] Hardware name: rockchip,rk3399-firefly-mini (DT)
[ 2.804445] task: ffffffc07187d400 ti: ffffffc071910000 task.ti:
ffffffc071910000
[ 2.805108] PC is at nvme_kthread+0x84/0x1f8
[ 2.805484] LR is at nvme_kthread+0x60/0x1f8
```

The conclusion is that the PCIE_AVDD_0V9 power supply is insufficient or unstable; try adding capacitors or increasing the voltage.